

Promise API

`Promise` class ke 6 static methods hain.

1. `Promise.all`

Kai promises **parallel** chalao aur **sabke poore hone** ka wait karo.

javascript

```
Promise.all([p1, p2, p3]).then(results => ...);
```

- Result ek array hota hai, aur order **wahi rehta hai jo input me tha** (chahe koi pehle khatam ho).
- Trick: data ko `map` karke promises ki array banao.

javascript

```
let requests = urls.map(url => fetch(url));  
Promise.all(requests).then(responses => ...);
```

- **Koi ek bhi reject** hua to `Promise.all` turant reject ho jaata hai, baaki ke results ignore ho jaate hain (par wo requests cancel nahi hoti).
- Array me normal values (promise nahi) bhi de sakte ho, wo jaise hain waise result me aa jaati hain.

2. `Promise.allSettled`

Sab ke **settle hone** ka wait karta hai, chahe success ho ya fail. Result me har ek ka status milta hai:

- `{ status: "fulfilled", value }`
- `{ status: "rejected", reason }`

Kaam ka: jab kuch requests fail ho jaayein tab bhi baaki chahiye ho.

javascript

```
Promise.allSettled(urls.map(u => fetch(u))).then(results => {  
  results.forEach((r, i) => {  
    if (r.status === "fulfilled") console.log(urls[i], r.value.status);  
    else console.log(urls[i], r.reason);  
  });  
});
```

(Purane browsers ke liye polyfill bana sakte ho.)

3. `Promise.race`

Sabse pehle settle hone wale promise ka result (ya error) leta hai. Baaki ignore.

```
javascript
```

```
Promise.race([slowP, fastP]).then(...); // fastP ka result
```

4. `Promise.any`

Sabse pehle fulfill hone wale promise ka result leta hai. Reject hone wale promises ko skip karta hai. Agar **saare reject** ho jaayein to `AggregateError` deta hai, jiske `errors` property me saare errors hote hain.

5. `Promise.resolve(value)`

Ready value se resolved promise banata hai. Jab function ko hamesha promise return karna ho (jaise cache se turant value dena):

```
javascript
```

```
function loadCached(url) {  
  if (cache.has(url)) return Promise.resolve(cache.get(url));  
  return fetch(url).then(r => r.text()).then(text => { cache.set(url, text); re  
}
```

6. `Promise.reject(error)`

Rejected promise banata hai. Practice me bahut kam use hota hai. (Aajkal `async/await` ki wajah se `resolve/reject` methods ki zarurat kam padti hai.)

Summary table

Method	Kab khatam hota hai	Result
<code>all</code>	sab fulfill hon	results ki array (ek fail to poorai fail)
<code>allSettled</code>	sab settle hon	status objects ki array
<code>race</code>	pehla settle	uska result/error
<code>any</code>	pehla fulfill	uska result (sab fail to <code>AggregateError</code>)
<code>resolve</code>	turant	value ke saath resolved
<code>reject</code>	turant	error ke saath rejected

Inme se `Promise.all` sabse zyada use hota hai.